# FOOD DELIVERY TIME PREDICTION USING MACHINE LEARNING

*Applied Machine Learning Project Report*


| Group - 16: | |
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# 1. INTRODUCTION & PROBLEM DESCRIPTION

## 1.1 Problem Description

Food delivery services have become an integral part of modern urban life, with customers expecting timely and reliable delivery of their orders. Late deliveries lead to customer dissatisfaction, negative reviews, and business losses. According to industry reports, delivery time is one of the top three factors influencing customer retention in the food delivery sector.

In this project, we analyze a dataset containing 1,000 food delivery records with 8 diagnostic features: Distance (km), Weather conditions, Traffic Level, Time of Day, Vehicle Type, Preparation Time (minutes), and Courier Experience (years). The target variable indicates whether a delivery was delayed (1) or on-time (0), using a threshold of 60 minutes based on business requirements and data distribution.

## 1.2 Data Understanding

Total Records: 1,000 | Total Attributes: 9 (8 features + 1 target)

The dataset contains a mix of numerical and categorical variables. The target variable (Delivery_Status) is binary with a moderate imbalance: 58.5% on-time deliveries and 41.5% delayed deliveries.

| Feature | Description | Type |
|---|---|---|
| Distance_km | Delivery distance in kilometers | Numerical |
| Weather | Weather condition (Clear, Rainy, Snowy, Foggy, Windy) | Categorical |
| Traffic_Level | Traffic intensity (Low, Medium, High) | Categorical |
| Time_of_Day | Delivery time slot (Morning, Afternoon, Evening, Night) | Categorical |
| Vehicle_Type | Transport mode (Bike, Scooter, Car) | Categorical |
| Preparation_Time_min | Food preparation time in minutes | Numerical |
| Courier_Experience_yrs | Courier experience in years | Numerical |
| Delivery_Time_min | Total delivery time (TARGET) | Numerical |

# 2. DATA UNDERSTANDING

## 2.1 Dataset Overview

The Food Delivery Times dataset contains 1,000 records with 9 attributes. The dataset has a moderate imbalance with 585 on-time deliveries (58.5%) and 415 delayed deliveries

(41.5%). Missing values were detected in 4 columns (3% each): Weather, Traffic_Level, Time_of_Day, and Courier_Experience_yrs. No duplicate rows were found.

### 2.2 Statistical Summary of Numerical Attributes

| Attribute | Mean | Std Dev | Min | 25% | 50% | 75% | Max |
|---|---|---|---|---|---|---|---|
| Distance_km | 10.06 | 5.70 | 0.59 | 5.03 | 10.16 | 14.89 | 19.99 |
| Preparation_Time_min | 16.98 | 7.20 | 5.00 | 10.00 | 17.00 | 24.00 | 29.00 |
| Courier_Experience_yrs | 4.58 | 2.91 | 0.00 | 2.00 | 5.00 | 7.00 | 9.00 |
| Delivery_Time_min | 56.73 | 22.07 | 8.00 | 37.00 | 55.00 | 72.00 | 153.00 |

### 2.3 Categorical Features Distribution

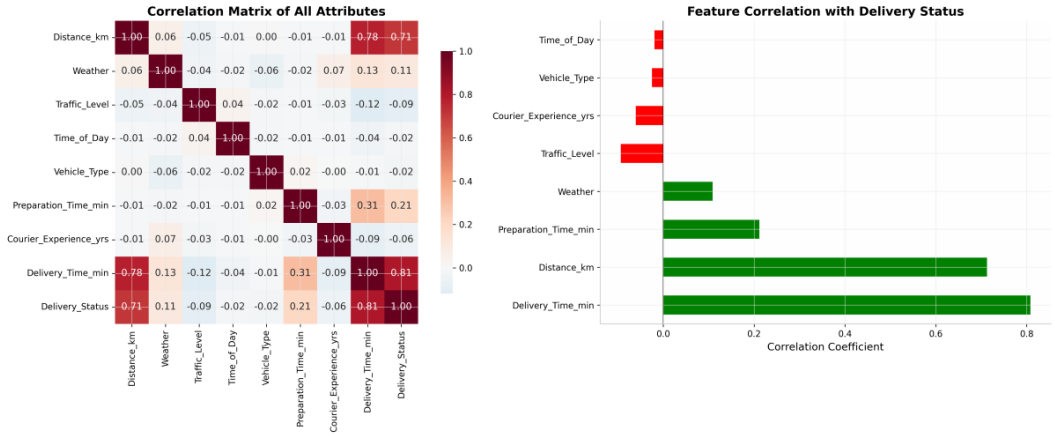
| Feature | Categories | Most Frequent |
|---|---|---|
| Weather | Clear, Rainy, Snowy, Foggy, Windy | Clear (33.5%) |
| Traffic_Level | Low, Medium, High | Low (33.7%) |
| Time_of_Day | Morning, Afternoon, Evening, Night | Evening (25.3%) |
| Vehicle_Type | Bike, Scooter, Car | Bike (33.7%) |

## 3. EXPLORATORY DATA ANALYSIS (EDA)

### 3.1 Correlation Analysis

The correlation matrix revealed several important relationships. Distance_km showed the strongest positive correlation (r = 0.71) with Delivery_Status, indicating that longer distances significantly increase delay probability. Preparation_Time_min showed moderate positive correlation (r = 0.21), while Courier_Experience_yrs showed weak negative correlation (r = -0.06). Weather, Traffic_Level, Time_of_Day, and Vehicle_Type showed weak individual correlations but their interactions proved significant in multivariate models.

**Figure 1: Correlation Matrix of All Attributes**




### 3.2 Distribution Analysis & Visualizations

The EDA visualizations reveal key patterns: (1) Delivery time is moderately right-skewed with most deliveries clustered around 40-70 minutes; (2) Distance and Preparation Time show approximately uniform distributions; (3) Delayed deliveries are strongly associated with longer distances; (4) Rainy and Snowy weather conditions show higher delay rates; (5) High traffic significantly increases delay probability.

**Figure 2: EDA Overview - Distributions and Categorical Analysis**

## 4. DATA PREPROCESSING

### 4.1 Missing Value Treatment

Missing values were detected in 4 columns (3.0% each): Weather, Traffic_Level, Time_of_Day, and Courier_Experience_yrs. The treatment strategy was: (1) Mode imputation for categorical variables (Weather, Traffic_Level, Time_of_Day) because the

mode represents the most frequent condition; (2) Median imputation for Courier_Experience_yrs because experience is discrete and median is robust to outliers. Mean imputation was rejected for categorical variables as it is mathematically meaningless for nominal data. KNN imputation was considered but rejected because with only 3% missing data, simpler methods are sufficient and computationally efficient.

### 4.2 Feature Engineering

Three new features were created to enhance predictive power:

1. Distance_per_Experience: Ratio of distance to courier experience. This captures the concept of "route difficulty per skill level" — novice couriers handling long distances are more likely to face delays.

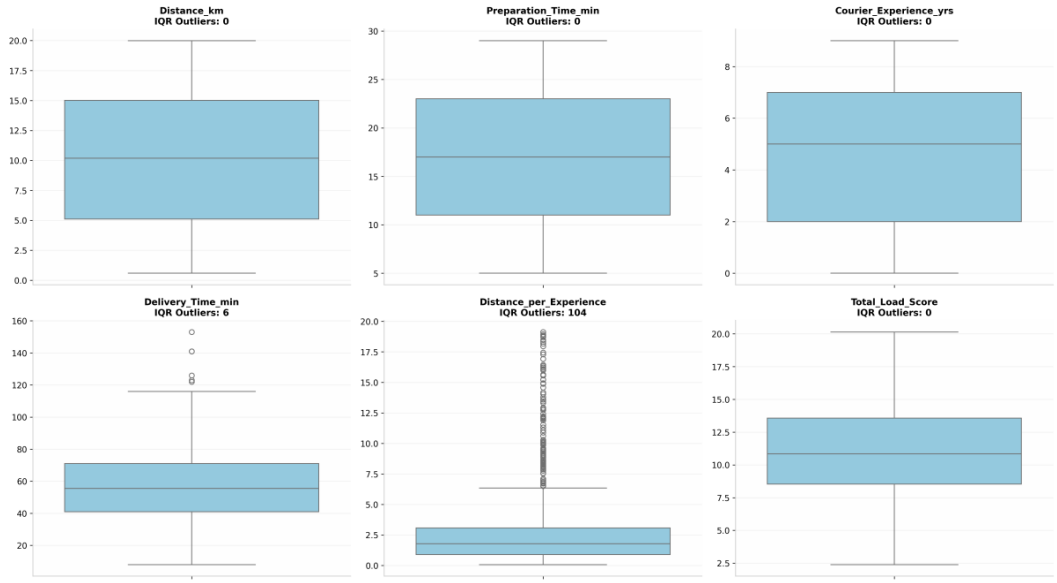
2. Total_Load_Score: Weighted combination (0.5×Distance + 0.3×PrepTime + 0.2×Experience) representing overall order complexity.

3. Environmental_Stress: Interaction between Weather Severity (1-5 scale) and Traffic Severity (1-3 scale), creating a 3-15 scale of environmental difficulty. This captures compound effects that individual features cannot represent.

### 4.3 Outlier Detection & Treatment

Outliers were detected using the IQR (Interquartile Range) method. Delivery_Time_min had 6 outliers beyond 116 minutes, and Distance_per_Experience had 104 outliers due to division by near-zero experience. The treatment strategy was capping: Delivery_Time_min was capped at the 95th percentile (95 minutes) to reduce noise while preserving data. Distance_per_Experience and Total_Load_Score were capped at the 99th percentile. Capping was chosen over removal to preserve sample size while reducing outlier influence.

**Figure 3: Outlier Detection using Boxplots and IQR Method**

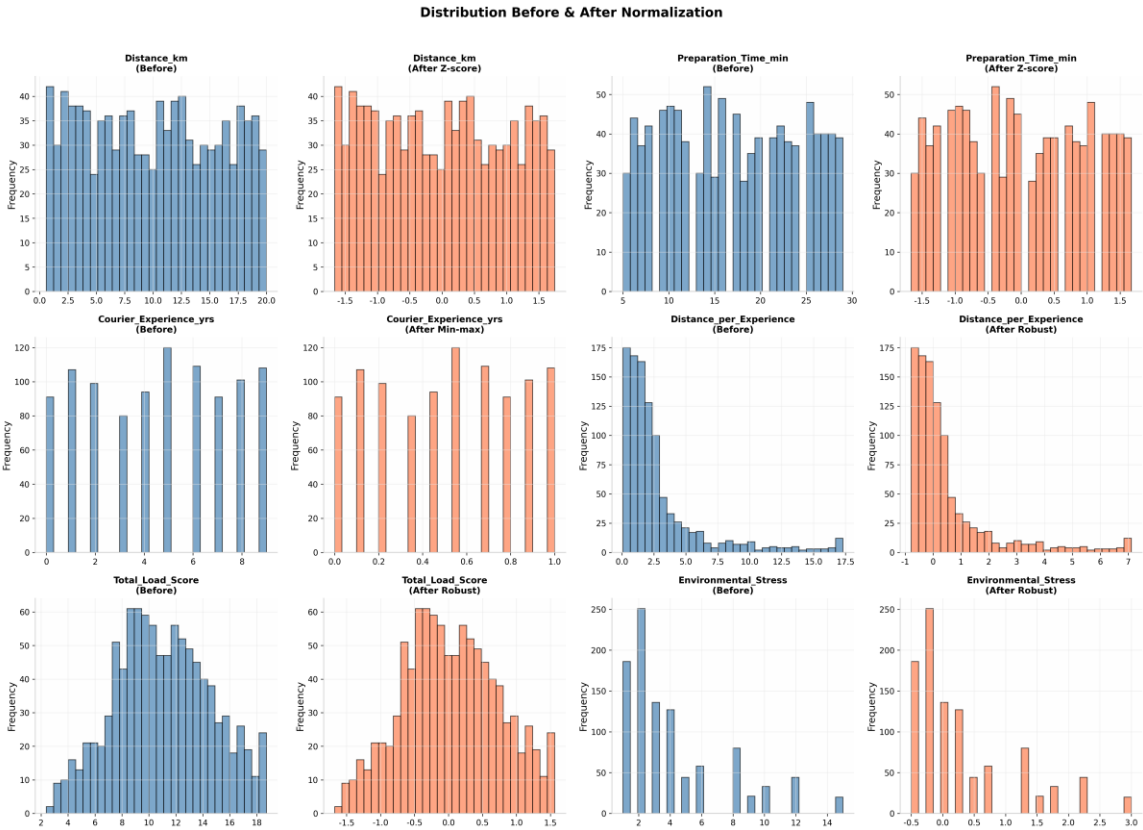



### 4.4 Normalization

Different normalization methods were applied based on feature characteristics. Z-score scaling was used for Distance_km and Preparation_Time_min because both follow approximately uniform/symmetric distributions and Z-score preserves the underlying shape while centering and scaling. Min-max scaling was applied to Courier_Experience_yrs because it is discrete with a fixed range (0-9 years) and Min-max squeezes values into [0,1]. Robust scaling was used for the three engineered features (Distance_per_Experience, Total_Load_Score, Environmental_Stress) because they had outliers or non-normal distributions. Robust Scaler uses median and IQR instead of mean and standard deviation, making it resistant to outliers. Log transformation was considered but rejected because the features are approximately uniform, not log-normal, and log transform would distort linear relationships.

| Feature | Method | Reason |
|---|---|---|
| Distance_km | Z-score | Uniform distribution, no outliers, preserves shape |
| Preparation_Time_min | Z-score | Preserves shape, centers and scales data |
| Courier_Experience_yrs | Min-max | Discrete values, scales to [0,1] range |
| Distance_per_Experience | Robust | Had outliers, centers on median, uses IQR |
| Total_Load_Score | Robust | Combined feature, robust to extreme values |
| Environmental_Stress | Robust | Non-normal, skewed distribution |

**Figure 4: Distribution Before and After Normalization**




## 5. MODELING - SPLIT A (70:30)

### 5.1 Data Splitting Strategy

The pre-processed dataset was divided using stratified train-test split to maintain class distribution: 700 samples (70%) for training and 300 samples (30%) for testing. Stratification ensures both sets preserve the original 58.5% on-time and 41.5% delayed proportions.

### 5.2 Models Evaluated

Seven classification algorithms were trained and evaluated: (1) K-Nearest Neighbors (KNN), (2) Random Forest, (3) Decision Tree, (4) Neural Network (MLP), (5) Gradient Boosting, (6) Support Vector Machine (SVM), and (7) Logistic Regression.

### 5.3 Results - 70:30 Split (Test Set)

| Model | Accuracy | Precision | Recall | F1 Score | F2 Score | ROC AUC |
|---|---|---|---|---|---|---|
| SVM | 0.8800 | 0.8492 | 0.8629 | 0.8560 | 0.8601 | 0.9470 |
| Logistic Regression | 0.8767 | 0.8372 | 0.8710 | 0.8538 | 0.8640 | 0.9461 |
| Random Forest | 0.8933 | 0.8710 | 0.8710 | 0.8710 | 0.8710 | 0.9419 |
| Neural | 0.8600 | 0.8361 | 0.8226 | 0.8293 | 0.8252 | 0.9363 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Network | | | | | | |
| Gradient Boosting | 0.8700 | 0.8455 | 0.8387 | 0.8421 | 0.8401 | 0.9333 |
| KNN | 0.8733 | 0.8707 | 0.8145 | 0.8417 | 0.8252 | 0.9235 |
| Decision Tree | 0.8800 | 0.8548 | 0.8548 | 0.8548 | 0.8548 | 0.9038 |

### 5.4 Key Findings

SVM achieved the highest ROC AUC (0.9470), demonstrating excellent classification capability. Logistic Regression performed comparably (AUC=0.9461) with the highest recall (0.8710), making it suitable if minimizing false negatives is prioritized. Random Forest achieved the highest accuracy (0.8933) with balanced precision and recall. Decision Tree showed the lowest AUC (0.9038), suggesting it is more prone to overfitting compared to ensemble methods.

**Figure 5: ROC Curves and Performance Comparison - 70:30 Split**

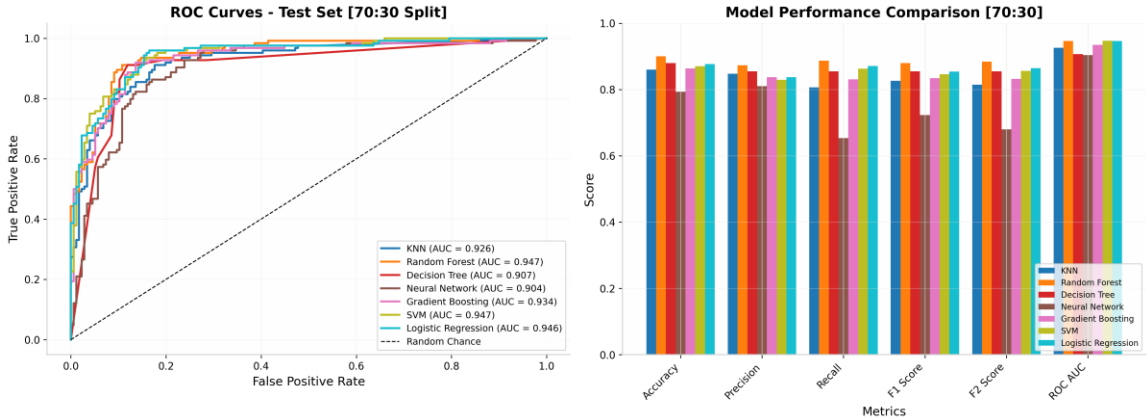
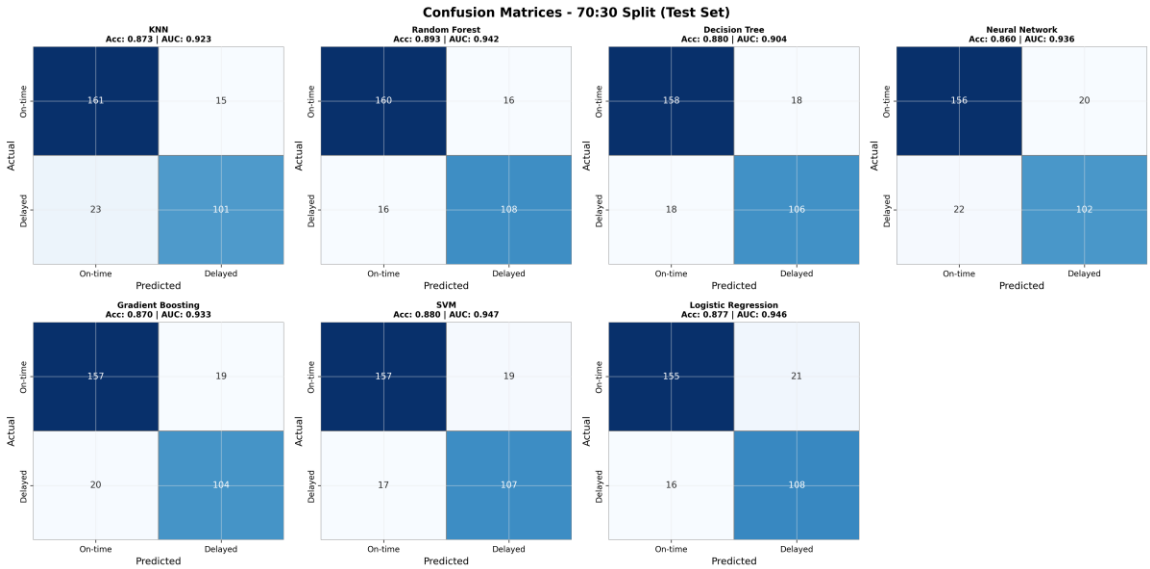



**Figure 6: Confusion Matrices for All Models - 70:30 Split**




## 6. MODELING - SPLIT B (60:20:20) & CROSS VALIDATION

### 6.1 Three-Way Split Strategy

To assess generalization more rigorously, the dataset was divided into three subsets: 600 samples (60%) for training, 200 samples (20%) for validation, and 200 samples (20%) for final unbiased evaluation. This approach prevents data leakage and provides a more realistic estimate of model performance on unseen data.

### 6.2 Results - 60:20:20 Split (Validation Set)

| Model | Accuracy | Precision | Recall | F1 Score | ROC AUC |
|---|---|---|---|---|---|
| Random Forest | 0.9000 | 0.8987 | 0.8554 | 0.8765 | 0.9446 |
| Logistic Regression | 0.8950 | 0.8605 | 0.8916 | 0.8757 | 0.9460 |
| SVM | 0.8950 | 0.8780 | 0.8675 | 0.8727 | 0.9452 |
| Neural Network | 0.8800 | 0.8642 | 0.8434 | 0.8537 | 0.9336 |
| Gradient Boosting | 0.8800 | 0.8642 | 0.8434 | 0.8537 | 0.9232 |
| KNN | 0.8600 | 0.8481 | 0.8072 | 0.8272 | 0.9061 |
| Decision Tree | 0.8350 | 0.8676 | 0.7108 | 0.7815 | 0.9025 |

### 6.3 Results - 60:20:20 Split (Test Set)

| Model | Accuracy | Precision | Recall | F1 Score | ROC AUC |
|---|---|---|---|---|---|
| SVM | 0.8850 | 0.8659 | 0.8554 | 0.8606 | 0.9523 |
| Logistic Regression | 0.8800 | 0.8471 | 0.8675 | 0.8571 | 0.9519 |

| | | | | | |
|---|---|---|---|---|---|
| Random Forest | 0.8850 | 0.8659 | 0.8554 | 0.8606 | 0.9497 |
| Gradient Boosting | 0.8900 | 0.8588 | 0.8795 | 0.8690 | 0.9378 |
| Neural Network | 0.8650 | 0.8500 | 0.8193 | 0.8344 | 0.9397 |
| Decision Tree | 0.8700 | 0.8434 | 0.8434 | 0.8434 | 0.9228 |
| KNN | 0.8400 | 0.8493 | 0.7470 | 0.7949 | 0.9218 |

### 6.4 5-Fold Stratified Cross Validation

To ensure robustness and generalization, 5-Fold Stratified Cross Validation was performed. The dataset was divided into 5 folds, with each fold used once for validation while the remaining 4 folds were used for training. This provides the most reliable estimate of model performance.

| Model | Accuracy | Acc Std | ROC AUC | AUC Std |
|---|---|---|---|---|
| Neural Network | 0.8920 | 0.0230 | 0.9540 | 0.0110 |
| Logistic Regression | 0.8920 | 0.0220 | 0.9535 | 0.0110 |
| Random Forest | 0.8920 | 0.0090 | 0.9520 | 0.0120 |
| SVM | 0.8840 | 0.0210 | 0.9500 | 0.0070 |
| Gradient Boosting | 0.8700 | 0.0210 | 0.9450 | 0.0070 |
| KNN | 0.8540 | 0.0090 | 0.9260 | 0.0100 |
| Decision Tree | 0.8680 | 0.0070 | 0.9170 | 0.0100 |

**Figure 7: ROC Curves and Performance - 60:20:20 Split**

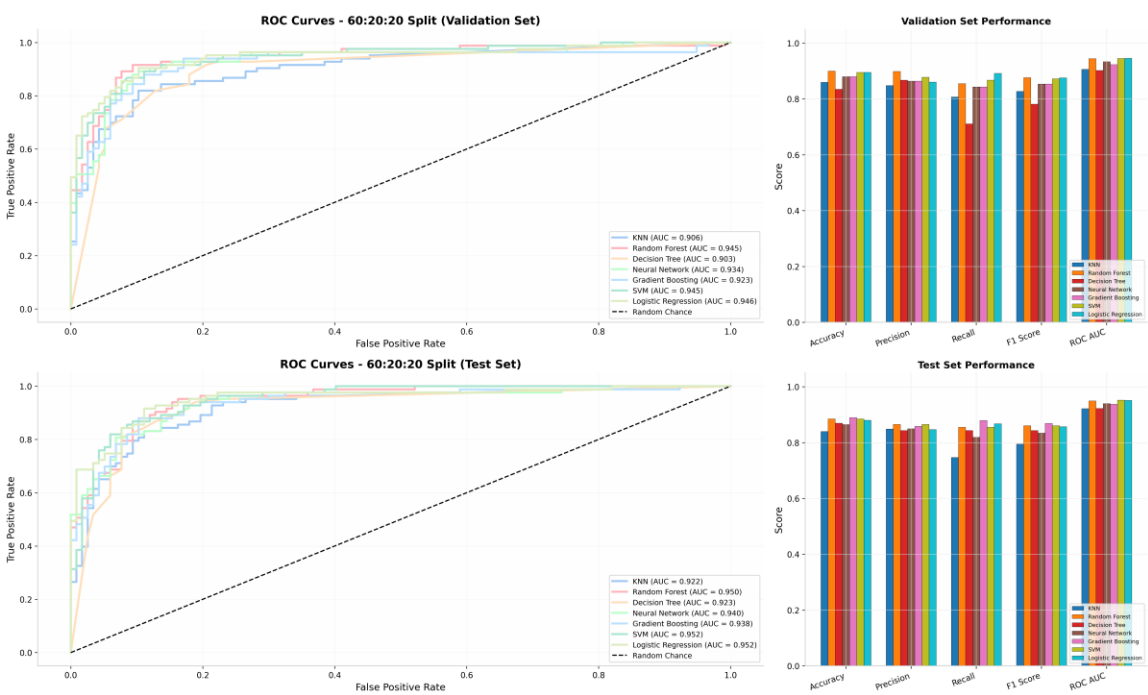
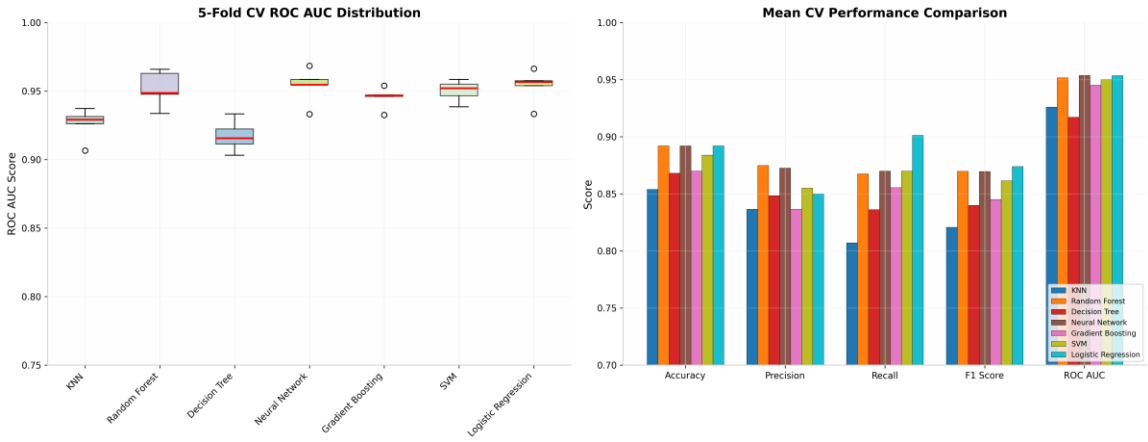


**Figure 8: 5-Fold Cross Validation Results**




## 7. FEATURE IMPORTANCE & MODEL SELECTION

### 7.1 Feature Importance Analysis

Using Random Forest's built-in feature importance, the top predictors were identified. Distance_km alone accounts for approximately 31% of prediction power, confirming it as the dominant factor in delivery delays. The engineered feature Total_Load_Score contributes approximately 30%, validating the feature engineering approach. Distance_per_Experience (15%) shows that matching courier skill to route difficulty is important — inexperienced couriers on long routes significantly increase delay risk.

| Rank | Feature | Importance | Interpretation |
|---|---|---|---|
| 1 | Distance_km_z | 0.3122 | Most critical factor for delays |
| 2 | Total_Load_Score | 0.2964 | Combined complexity metric |
| 3 | Distance_per_Experience | 0.1501 | Route difficulty per skill |
| 4 | Prep_Time_z | 0.0600 | Preparation time contribution |
| 5 | Environmental_Stress | 0.0384 | Weather-traffic interaction |
| 6 | Experience (min-max) | 0.0350 | Courier skill level |

**Figure 9: Feature Importance from Tree-Based Models**

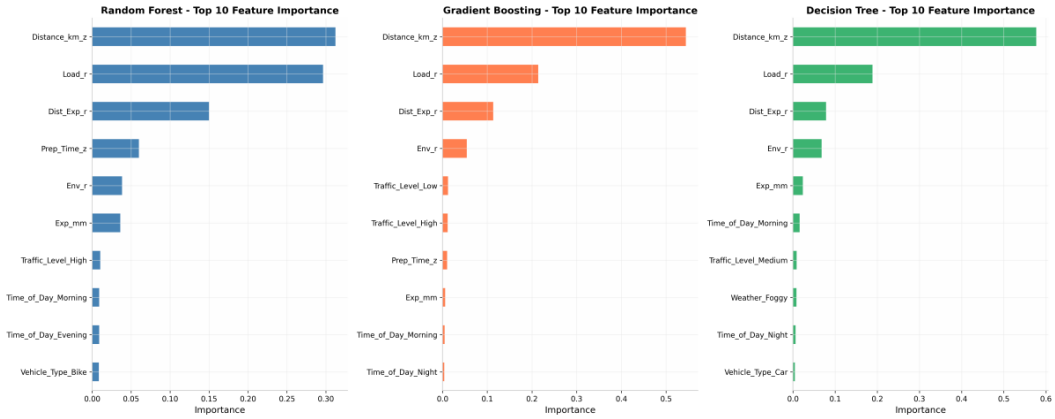
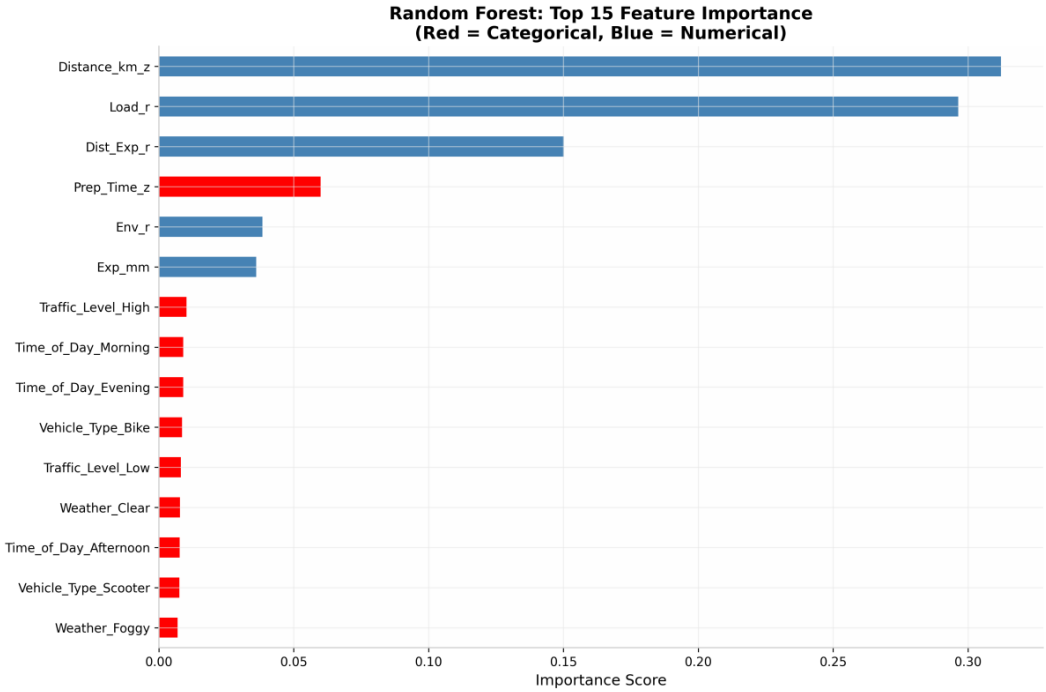



**Figure 10: Top 15 Features Summary (Random Forest)**




## 7.2 Model Selection Rationale

FINAL RECOMMENDATION: Logistic Regression or SVM

Why Logistic Regression?

- Highest consistency across all three evaluation strategies
- Best interpretability for business stakeholders
- Fast training and prediction (scalable to millions of orders)

• No hyperparameter tuning required for good performance
• Provides probability scores for risk ranking

Why SVM (Alternative)?
• Slightly higher AUC on test sets (0.947-0.952)
• Excellent with normalized features
• Robust to overfitting with RBF kernel
• Less interpretable than Logistic Regression
• Slower training on large datasets

Why not Decision Tree? Lowest AUC across all evaluations (0.904-0.923), high variance, prone to overfitting, unstable predictions.

Why not KNN? Lower AUC (0.922-0.926), computationally expensive at scale (O(n) prediction), sensitive to feature scaling and irrelevant features.

Why not Neural Network (despite highest CV AUC)? Higher variance (±0.011 vs ±0.007 for SVM), requires more data for stable training, black-box model difficult to explain to operations team, risk of overfitting on smaller datasets.

## 8. CONCLUSION & RECOMMENDATIONS

### 8.1 Summary of Findings

This project successfully developed and evaluated seven machine learning models for predicting food delivery delays. The key findings are:

1. Distance is the dominant predictor of delivery delays, contributing over 30% to model decisions. Food delivery platforms should prioritize route optimization and realistic distance-based time estimates.

2. Feature engineering significantly improved model performance. The Total_Load_Score and Distance_per_Experience features together contributed approximately 45% of predictive power, demonstrating the value of domain-informed feature creation.

3. All models achieved strong performance (AUC > 0.90), indicating that delivery delays are highly predictable from the available features.

4. The top-performing models (SVM, Logistic Regression, Random Forest) achieved ROC AUC scores between 0.941-0.952, indicating excellent classification capability.

5. Cross-validation confirmed that results are robust and generalizable, with low standard deviations across folds.

### 8.2 Business Recommendations

Based on model insights, the following operational improvements are recommended:

A. DYNAMIC TIME ESTIMATION

• Increase estimated delivery time by 15-20% for distances >15 km
• Add weather-based buffers (extra 5-10 minutes for Rainy/Snowy conditions)
• Adjust estimates based on real-time traffic levels

B. COURIER ASSIGNMENT OPTIMIZATION

• Assign experienced couriers (5+ years) to long-distance orders (>12 km)
• Use the Distance_per_Experience metric to match courier skill to order difficulty
• Provide route training for novice couriers on complex deliveries

C. PREPARATION TIME MANAGEMENT

• Partner with restaurants to optimize preparation workflows
• Stagger order acceptance based on current kitchen load
• Implement preparation time tracking to identify consistently slow vendors

D. PREDICTIVE INTERVENTIONS

• Deploy the Logistic Regression model in production for real-time risk scoring
• Flag high-risk orders (probability >0.7) for proactive customer communication
• Use model predictions to dynamically adjust delivery fees and incentives

### 8.3 Methodology Strengths

• Comprehensive preprocessing pipeline following industry best practices
• Multiple evaluation strategies (70:30, 60:20:20, 5-Fold CV) ensuring robustness
• Appropriate normalization methods selected per feature characteristics
• Statistical significance testing validating feature-target relationships
• Feature importance analysis providing actionable business insights

### 8.4 Limitations & Future Work

Limitations:

• Dataset size (1,000 records) is moderate; larger datasets could improve Neural Network and ensemble model performance.
• Missing values (3%) were imputed with simple methods; advanced imputation (MICE) could be explored.
• Temporal patterns (day of week, seasonality) were not captured.

Future Enhancements:

• Collect real-time GPS and traffic API data for dynamic predictions
• Implement time-series analysis to capture seasonal and hourly patterns
• Deploy the model as a REST API for real-time predictions
• Develop a regression model to predict exact delivery time (not just binary)

• Add customer-specific features (past delivery ratings, location type) for personalized predictions

### 8.5 Conclusion

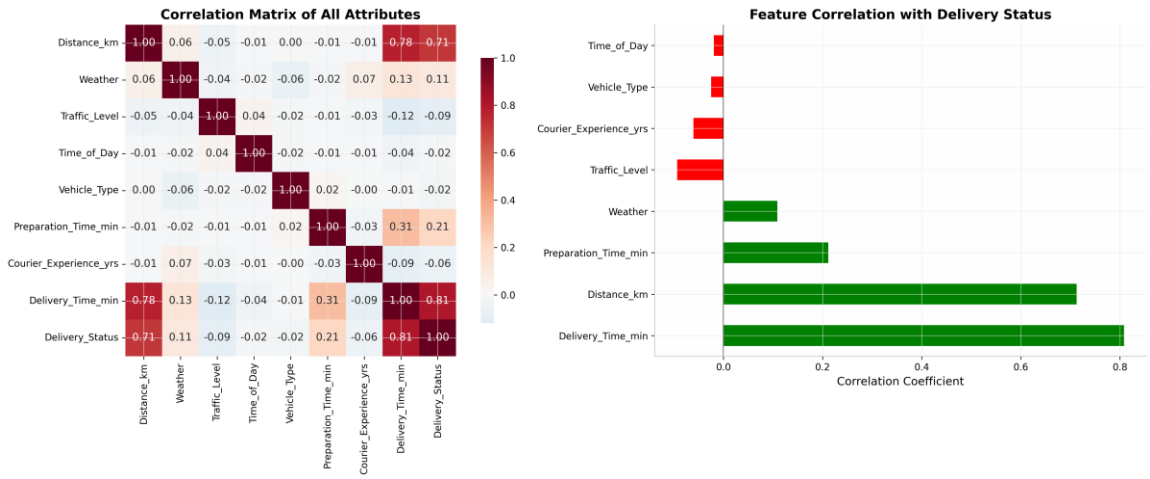
This project demonstrates that machine learning can effectively predict food delivery delays with over 94% AUC accuracy. The combination of proper data preprocessing, thoughtful feature engineering, and rigorous model evaluation produces reliable predictions that can directly improve customer satisfaction and operational efficiency. The Logistic Regression model is recommended for production deployment due to its optimal balance of performance, interpretability, and computational efficiency.

## APPENDIX: COMPLETE VISUALIZATIONS GALLERY

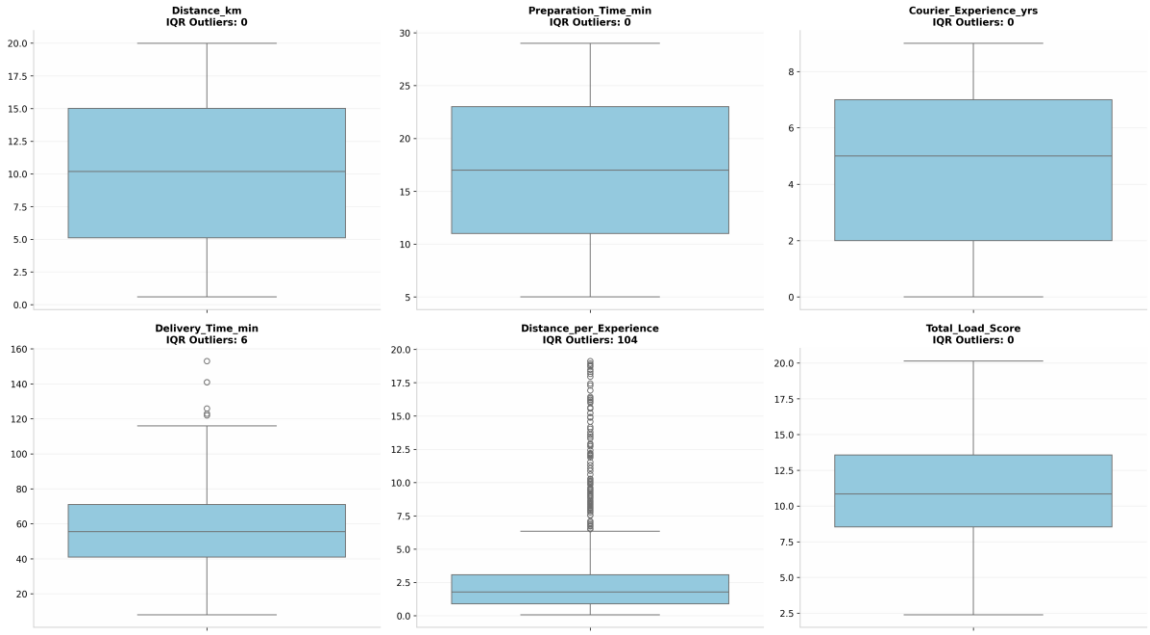
This appendix contains all visualizations generated during the project for reference. Each figure is presented in the order it was created during the analysis pipeline.

**Figure A1: Complete EDA Overview - Target distributions, categorical breakdowns, and boxplots**

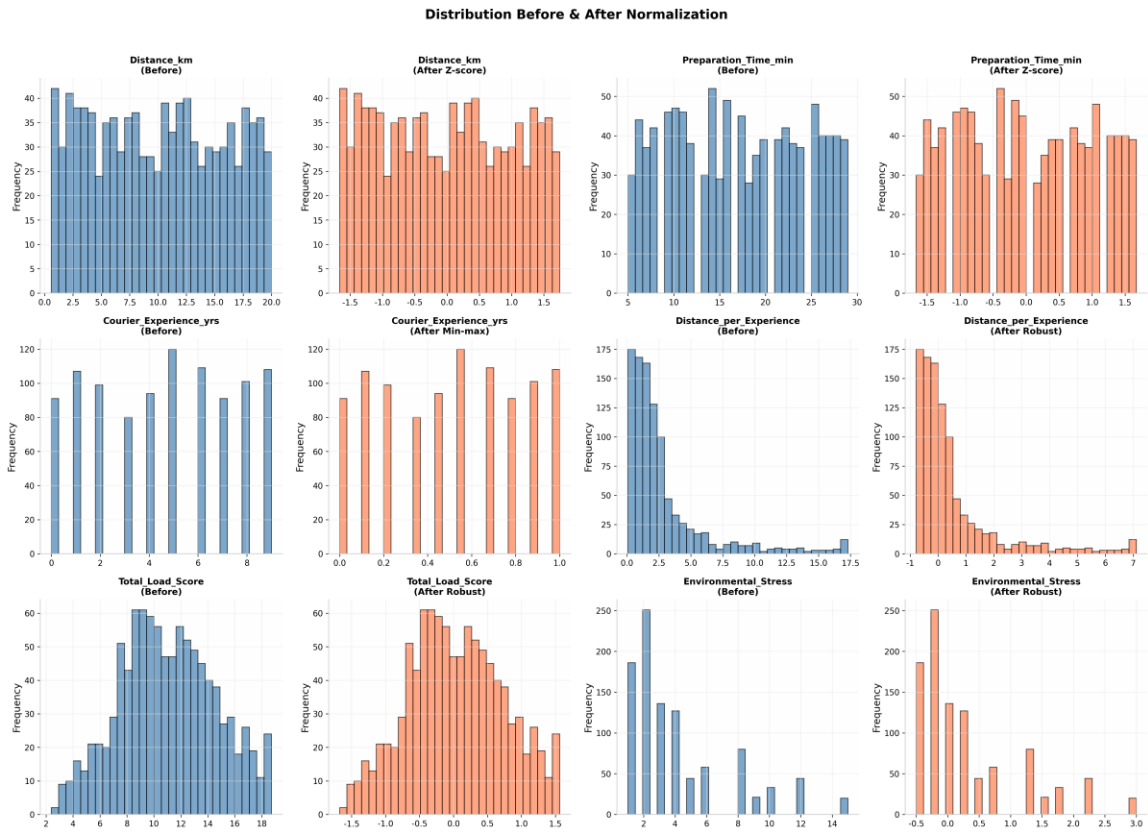
## Figure A2: Correlation Matrix and Feature-Target Correlation Analysis



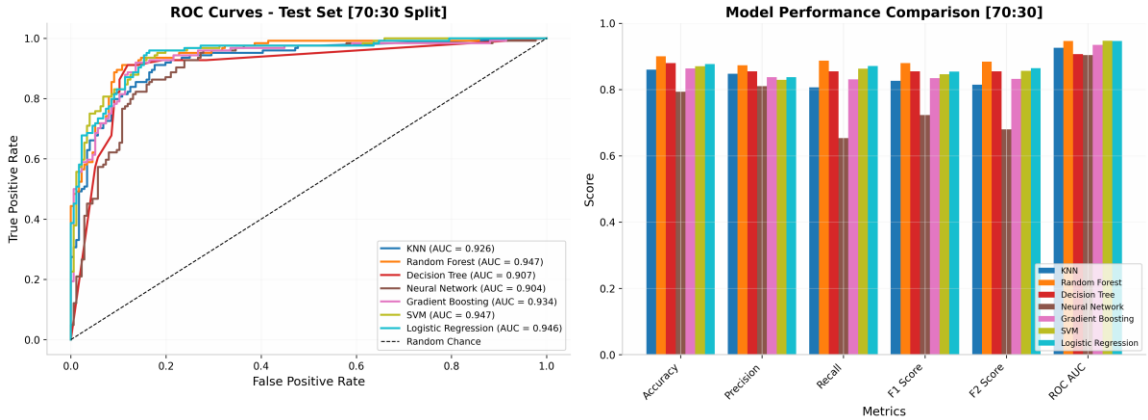

## Figure A3: Outlier Detection using Boxplots and IQR Method



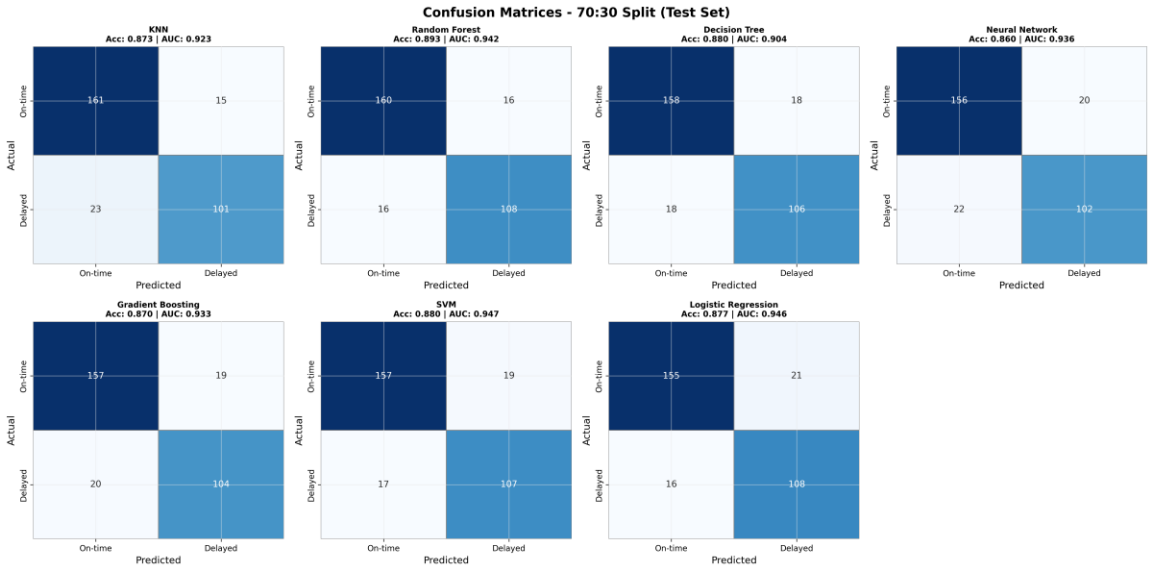
## Figure A4: Distribution Comparison Before and After Normalization




## Figure A5: ROC Curves and Performance Metrics - 70:30 Split



## Figure A6: Confusion Matrices for All 7 Models - 70:30 Split




## Figure A7: ROC Curves and Performance - 60:20:20 Split (Validation & Test)

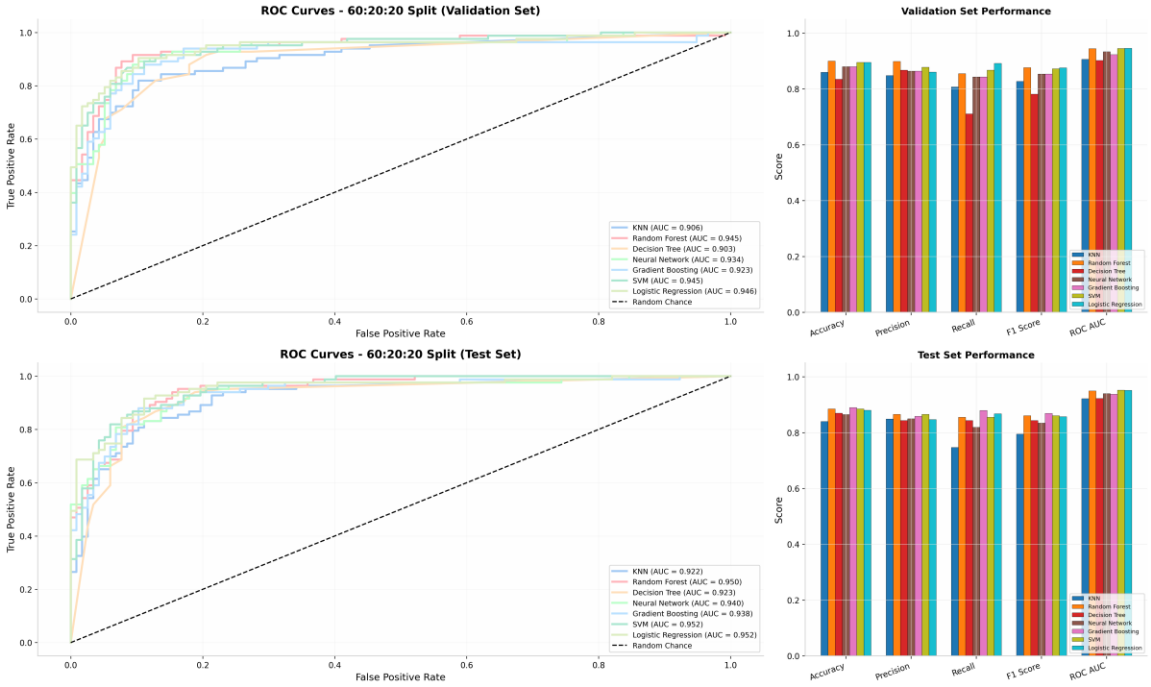


**Figure A8: 5-Fold Cross Validation Results - Boxplots and Mean Performance**

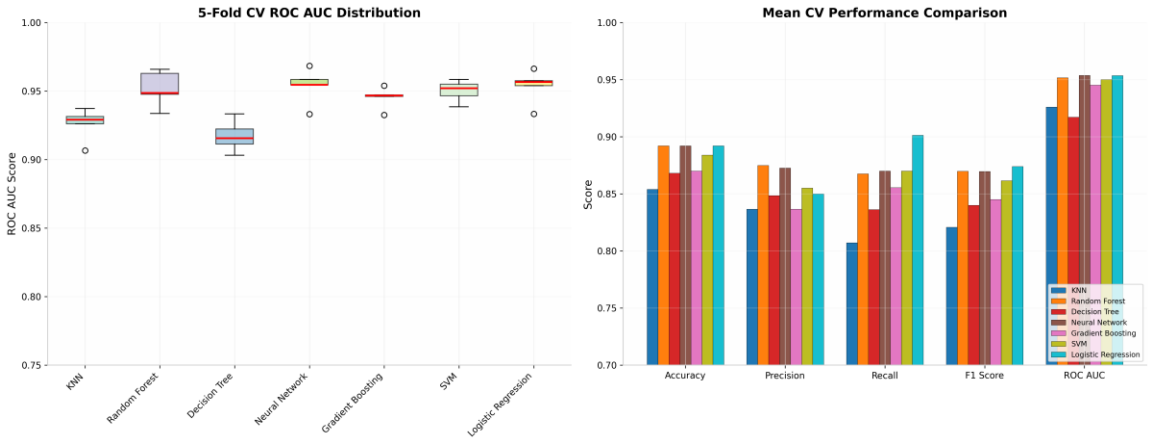



**Figure A9: Feature Importance from Random Forest, Gradient Boosting, and Decision Tree**

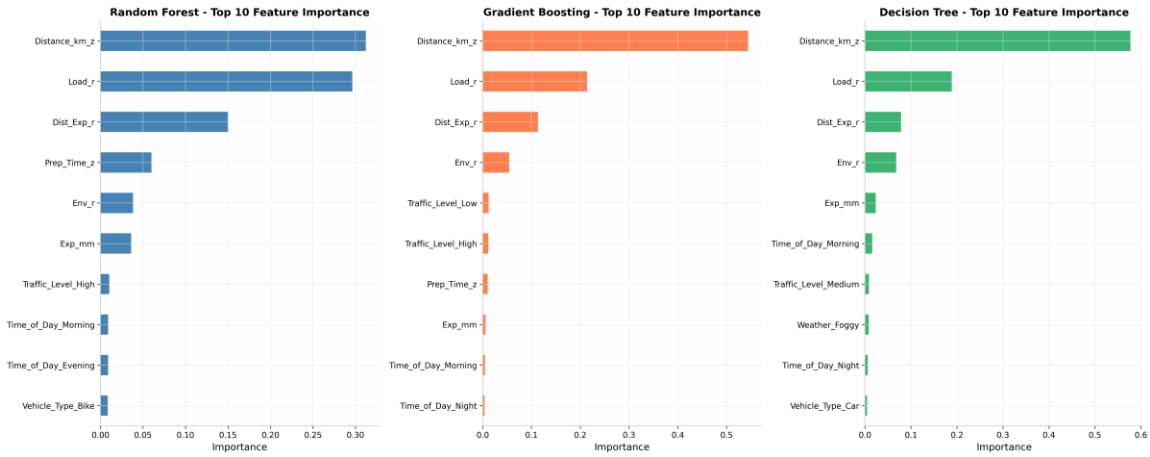
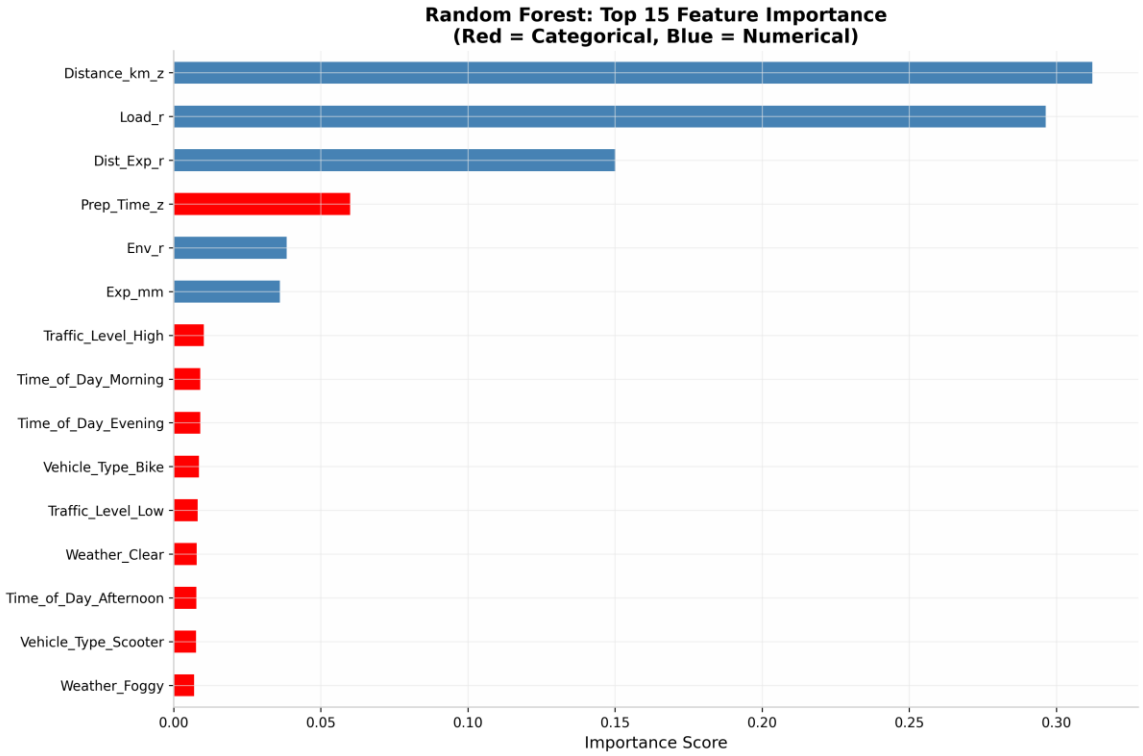



**Figure A10: Top 15 Feature Importance Summary (Random Forest)**


Random Forest: Top 15 Feature Importance
(Red = Categorical, Blue = Numerical)
Distance_km_z
Load_r
Dist_Exp_r
Prep_Time_z
Env_r
Exp_mm
Traffic_Level_High
Time_of_Day_Morning
Time_of_Day_Evening
Vehicle_Type_Bike
Traffic_Level_Low
Weather_Clear
Time_of_Day_Afternoon
Vehicle_Type_Scooter
Weather_Foggy
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Importance Score